

VISHAL KUMAR

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Education

University of Southern California

Master's of Science in Computer Science

2025 – 2027

Los Angeles, CA

Guru Gobind Singh Indraprastha University

B.Tech. in Computer Science

2020 – 2024

New Delhi, India

Experience

Teaching Assistant, USC Viterbi K-12 STEM Center

Jun 2025 – Aug 2025

University of Southern California

Los Angeles, CA

- Delivered 40+ hours of instruction in Cybersecurity, Python, and Game Development to 100+ high school students as part of USC's K-12 outreach per week.
- Designed structured learning modules and projects, increasing student task completion rates by **25%**.
- Guided students during labs and office hours, helping debug code and reinforce core CS concepts and programming fundamentals.

Software Developer Intern

Jun 2024 – Sep 2024

Punjab National Bank

New Delhi, India

- Developed and deployed a Litigation Management System using Java Servlets, JSP, and MySQL, streamlining case tracking and document workflows.
- Built RESTful APIs using Node.js and Express to automate banking processes, reducing processing time by **40%**.
- Optimized backend logic and database queries, improving response time by **50%** under ~500 concurrent users and reducing errors by **25%**.
- Deployed on Apache Tomcat and collaborated in Agile sprints, improving release efficiency by **15%**.

Research Publications

Generative A.I. Integration using APIs

2023

- Published in Satyam, official journal of GGSIPU, under guidance of Dr. Sangeeta and Dr. Sitender.
- Conducted literature-driven research on Large Language Model (LLM) and API integration strategies, focusing on reliability, scalability, and practical software patterns.
- Synthesized best practices for designing AI-augmented workflows without introducing new model architectures, emphasizing real-world applicability.

Technical Skills

Languages: Python, JavaScript, Go, Swift, SQL/MySQL, C#

ML & Data: TensorFlow, YOLOv8, NumPy, pandas, scikit-learn

Frameworks: FastAPI, Node.js, React, Next.js, Celery, Tailwind CSS

Databases: PostgreSQL, Redis, Firebase, MySQL

Tools: Docker, Claude Code, Cursor, Git/GitHub, Copilot, Hibernate, Base44, Huggingface

Projects

Transizr — AI-Powered Media Transcription Platform | *FastAPI, Celery, RabbitMQ, Redis, Docker* 2024 – Present

- Developed a distributed transcription system using containerized workers and message queues for scalable audio/video-to-text processing.
- Built Celery-RabbitMQ pipelines enabling parallel, fault-tolerant task execution across multiple containers.
- Added real-time job tracking using Redis Pub-Sub and Server-Sent Events (SSE) for instant UI updates.

SightRanger — AI Surveillance System | *FastAPI, YOLOv8, Next.js, Docker*

2024

- Implemented an intrusion-detection pipeline using YOLOv8 for real-time human detection from webcam/mobile streams.
- Built a Next.js dashboard to view detections, alerts, and clip history with streamlined monitoring tools.
- Containerized backend and frontend with Docker Compose for reproducible deployments and easy scaling.

CodeBundle — Directory-to-Text Export CLI | *Go, CLI*

2023

- Built a Go-based CLI that bundles entire codebases into a structured .txt file for LLM workflows and code review.
- Added a high-speed *Fast Mode* that optimizes file collection for large repositories.